IMU Data Analysis Report

Per-vendor IMU contracts (NovAtel, VectorNav, Point One Nav), empirical findings from run_2 mcap, and recommendation for race_common's INS-corrected IMU strategy.
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1. Executive summary

Empirical inspection of run_2 proves that /gps_na/imu and /gps_p1/imu both carry **specific-force IMU** (gravity present) with **no fused orientation**. They differ in one respect: the NovAtel topic is **raw** (RAWIMUSX, no sensor-level calibration applied), while the Point One Atlas topic is **sensor-calibrated** (imu_calibrated: bias, scale and misalignment removed by Atlas firmware). The VectorNav topic /gps_nav/imu is not present in run_2 but the iac_launch interface node is configured with imu.calibrated: true, which means the produced stream is sensor-calibrated specific force, analogous to /gps_p1/imu.

Bottom line for GICP: the current bias compromise — initial RTK/stationary calibration only, K_ab = K_gb = 0 — is the correct resting point. GICP's a[2] -= gravity_ integration logic is correct for all three vendor streams as currently published. We do **not** recommend switching to gravity-removed INS IMU; we *do* recommend a small, optional improvement on the NovAtel side to bring it in line with the other two vendors.

2. Method

Two evidence streams:

Empirical: read every sensor_msgs/Imu message from run_2_0.mcap for each available IMU topic. For each stream: mean / std of accel + gyro over the whole bag and over the first 30 s (presumed stationary), orientation populated check, covariance field uniqueness, frame_id, and rate. The verdict on "gravity present?" is read directly off a_z mean.

Driver-side: for each vendor, examine the launch + parameter files in airacingtech/race_common (branch stable) to confirm which raw or filtered log family is enabled and which interface node re-publishes it on the /gps_*/imu topic.

3. Per-vendor IMU contract

vendor	topic	source	bias/scale comp.	gravity removed?	fused orientation?	rate	frame_id
NovAtel PwrPak7D-E1 + Epson G320N	/gps_na/imu	RAWIMUSX via novatel_interface (driver receives ID 1462 only, no 2264 CORRIMUSX, no 813 IMURATECORRIMUS)	**NO** — raw counts scaled by IMU type	**NO** — specific force, $a_z \approx +9.81$ m/s^2	**NO** — orientation identically (1,0,0,0) across 205k samples	**198 Hz** (200 Hz nominal)	novatel_a
Point One Nav Atlas LG69T	/gps_p1/imu	Atlas imu_calibrated via pointonenav_interface	**YES** — sensor-level bias/scale/misalignment removed by Atlas firmware	**NO** — specific force, $a_z \approx +9.82$ m/s^2	**NO** — orientation identically (1,0,0,0) across 102k samples; orientation_covariance[0]=1000 (flagged as untrusted)	**99 Hz**	gps_antenna_top
VectorNav VN-300/310	/gps_nav/imu	VectorNav binary imu_group (/vectornav/raw/imu) via vectornav_interface; in iac_launch param: calibrated: true	**YES** (per calibrated: true flag) — VN firmware bias/scale comp.	Expected **NO** — VN compensated IMU emits specific force; gravity is only removed in ins_group linear-acceleration. Not present in run_2 bag for empirical verification.	Expected **NO** — orientation is populated only if the interface fuses attitude_group in	**not observed** (no /gps_nav/* topics in run_2)	*not observed*

4. Empirical numbers from run_2

metric	/gps_na/imu	/gps_p1/imu
samples	205,624	102,800
duration / rate	1036.4 s / 198.4 Hz	1035.5 s / 99.3 Hz
frame_id	novatel_a	gps_antenna_top
a_x mean / std (whole bag)	+0.046 / 0.526 m/s^2	+0.045 / 0.517 m/s^2
a_y mean / std	-0.114 / 0.845	+0.145 / 0.844
a_z mean / std	**+9.808** / 0.292	**+9.820** / 0.206

a_z first 30 s (stationary?)	+9.812 ± 0.004	+9.810 ± 0.003
ω std (rad/s, all axes)	0.009 / 0.008 / 0.094	0.007 / 0.007 / 0.094
orientation populated?	identity for all 205,624 samples	identity for all 102,800 samples
linear_acc_cov[0] distinct values	80,436 (varies per sample)	51,306 (varies per sample)
angular_vel_cov[0] distinct values	316 (~0 - 5×10^{-6})	4 (~0 - 5×10^{-8})
orientation_cov[0]	0.0 (mis-flagged as 'known')	1000.0 (explicit untrusted)

The two strongest tells: (a) a_z averages within 0.02 m/s^2 of standard gravity over a 17-minute drive on both topics, which only happens if linear_acceleration is specific force; (b) the orientation field is bit-identical to (w=1, x=y=z=0) for every sample, meaning no INS attitude is being injected. The static-window noise is <5 milli-g for both vendors — both Epson G320N (NovAtel) and Atlas's internal IMU are in good health.

5. Where the data actually comes from

5.1 NovAtel pipeline

The driver receives the following log IDs (per iac_launch/param/gps_param/novatel_a/std_oem7_raw_msgs.yaml): 42, 99, 1429, 1430, 1538, **1465 INSPVAX**, 1825, **1462 RAWIMUSX**, ephemerides, 43 RANGE, 1335 HEADING2, 101, 93, 1945 INSCONFIG, 1719, 1720, 1051, 1052.

CORRIMUSX (2264) and IMURATECORRIMUS (813) are not in the list. The upstream INSHandler::handleMsg only calls publishImuMsg() — the path that fills the IMU topic from corrimu_ + inspva_ caches — on receipt of CORRIMUSX, so this path never fires. Only processRawImuMsg runs, publishing the raw stream on RAWIMU (→ imu/data_raw in the upstream mapping). The novatel_interface node then ingests this raw stream (remap raw_imu → /novatel_a/imu/data_raw) and re-publishes it on its own imu output, remapped to /gps_na/imu. End result: /gps_na/imu is RAWIMUSX-derived. The empirical data agrees.

5.2 Point One Atlas pipeline

The fusion_engine_driver publishes imu_raw (raw), imu_calibrated (sensor-level bias/scale/misalignment removed by Atlas firmware), and the FusionEngine pose/INS solutions. The pointonenav_interface subscribes specifically to imu_calibrated and republishes it on its imu output, remapped to /gps_p1/imu.

Per FusionEngine semantics, imu_calibrated applies sensor-level corrections only — it does not subtract gravity, and it does not provide attitude. Atlas-fused attitude lives in pose_filtered / pose_aux, not on this IMU topic. So /gps_p1/imu is bias/scale-corrected specific force without orientation. The empirical data agrees: stationary a_z = +9.810 m/s^2, orientation identity, orientation_covariance[0] = 1000 (explicitly untrusted).

5.3 VectorNav pipeline

Two driver nodes are started: vectornav (binary log decoder) and vn_sensor_msgs (binary → ROS msg converter). They publish raw VN binary groups on /vectornav/raw/{imu,ins,attitude,gps,gps2}. The vectornav_interface node then subscribes to those and produces the /gps_nav/* family.

The key parameter is in iac_launch/param/gps_param/vectornav/vectornav_interface.param.yaml: imu.calibrated: true, with a rpy = [0, 0, -π/2] frame rotation. calibrated: true tells the interface to pull the VN-internal compensated IMU (bias and scale corrections applied by VN firmware). The rosbag and dspace launch contexts override this to calibrated: false — useful when replaying raw recordings without firmware corrections.

Per VectorNav's spec, the compensated IMU output is specific force; gravity-removed linear acceleration is only available in ins_group's AccelBody / AccelECEF fields, which this interface does not feed into /gps_nav/imu. So /gps_nav/imu, when active, would carry sensor-calibrated specific force with no fused orientation — analogous to /gps_p1/imu. We cannot empirically verify because run_2 does not contain /gps_nav/* topics; this is a confident prediction from the source, not an observation.

6. Should race_common publish INS-corrected IMU?

Short answer: **no, not in the "gravity-removed, INS-pose-fused" sense**. **Yes, in the "sensor-level bias/scale-corrected" sense for NovAtel, to match P1 and VectorNav**.

6.1 Why gravity-removed INS IMU is the wrong target

DLIO's IMU integration is built around specific force. integrateImu rotates raw lin_accel into world frame and subtracts gravity_; propagateState does the same in world Z. If the input stream had gravity removed by the vendor's INS, both subtractions would over-cancel and inject a phantom -1g into world Z. Removing them is doable but means structural code surgery and a new failure mode: the vendor's gravity removal depends on the vendor's INS attitude, which fails or drifts during alignment / RTK loss / aggressive maneuvers. Raw specific force has no such dependency.

Other downsides: vendor INS-corrected IMU is typically lower-rate (NovAtel CORRIMUSX runs at 100 Hz vs. RAWIMUSX 200 Hz) — halving the per-point deskew temporal resolution — and introduces an estimator feedback loop if any GICP output ever feeds back into the vendor's INS solution (currently it does not, but the team's VKS architecture creates the opportunity).

6.2 The one thing we DO recommend: NovAtel CORRIMUSX

The asymmetry in the current setup is purely accidental. P1 and VectorNav both expose *sensor-level calibrated* IMU through imu_calibrated / VN's compensated output. NovAtel could expose the same by enabling CORRIMUSX in std_oem7_raw_msgs.yaml, then either: (a) remap imu/data → /gps_na/imu and accept the 100 Hz rate; or (b) keep /gps_na/imu on RAWIMUSX (200 Hz) and add a parallel /gps_na/imu_calibrated topic so downstream consumers can choose. Sensor-level calibration removes static bias and scale-factor drift; it leaves gravity and specific-force semantics intact, so GICP needs no math change.

Stationary bench-test discriminator: with CORRIMUSX, stationary a_x/a_y should land in ±0.02 m/s^2 (gravity-of-tilt only). On the run_2 data, NovAtel's stationary a_x = +0.182 m/s^2 and a_y = -0.140 m/s^2 — that's ~1.8° equivalent tilt or ~18 mg of static bias. Atlas shows +0.171 m/s^2 / +0.072 m/s^2 on the same epoch — smaller, consistent with the Atlas-side sensor calibration already being applied.

6.3 What GICP should do today — no change

The recently merged compromise — odom/geo/Kab = odom/geo/Kgb = 0, initial RTK/stationary calibration retained — is correct for the current input contracts on all three vendors:

· specific-force IMU with possibly small residual bias (zero for P1/VN sensor-calibrated, MEMS-level for NovAtel raw) — needs *some* bias estimate, which initial calibration provides;

· gravity present — matches a[2] -= gravity_ in integrateImu and propagateState;

· no fused orientation — GICP does not consume orientation from sensor_msgs/Imu (verified: only angular_velocity and linear_acceleration are used). The identity-quaternion content is benign;

· bias-adaptation observer off — avoids the GICP residual chasing bias when the topic is already sensor-calibrated (P1 / VN) or when the stationary calibration has already absorbed the static MEMS bias (NovAtel).

7. Recommended follow-ups

Priority-ordered, smallest-effort first.

7.1 (small) Document the input contract per topic

Add a one-line comment in gicp_localization/cfg/localization.yaml next to imu_topic that states: 'Expects specific-force IMU (gravity present, sensor calibration may or may not be applied). Compatible with /gps_na/imu (raw RAWIMUSX), /gps_p1/imu (sensor-calibrated), /gps_nav/imu (sensor-calibrated). NOT compatible with gravity-removed INS IMU.' This is the single highest-value, lowest-effort intervention — it prevents a future regression where someone "upgrades" the topic to a gravity-removed source.

7.2 (small) Tighten clamps and add a freeze-after-init flag

abias_max = 5 m/s^2 and gbias_max = 0.5 rad/s are unreachable with K_ab = K_gb = 0, but if anyone re-enables the observer they're 50×100× larger than physical MEMS drift. Drop them to 0.5 / 0.05 so future re-enables don't soak extrinsic-mismatch error into bias.

7.3 (medium) Add the NovAtel CORRIMUSX option in race_common

In iac_launch/param/gps_param/novatel_a/std_oem7_raw_msgs.yaml add log ID 2264 (CORRIMUSX). The upstream INSHandler::publishImuMsg() will then emit sensor-calibrated specific force on imu/data. Decide between (a) remap imu/data → /gps_na/imu (loses 100 Hz of rate but symmetrizes vendors); (b) keep /gps_na/imu on raw, expose calibrated as /gps_na/imu_calibrated (parallel to Atlas's /atlas/imu_calibrated). Option (b) is the lower-regret choice — gives the team an A/B knob.

7.4 (medium) Log and surface vendor IMU clock skew

The 200 Hz NovAtel stream and 100 Hz Atlas stream do not align on the same epoch. GICP currently chooses one (/gps_na/imu) and ignores the other. If we ever want to cross-check or fuse, we'll need a one-shot diagnostic that captures the relative header.stamp offset and rate ratio. Patterned after the Luminar timestamp diagnostic already merged.

7.5 (medium) Make the bag recording include INS-corrected sensor outputs

The Atlas firmware already publishes /atlas/imu_calibrated; the VectorNav interface produces a full /gps_nav/* family. Neither shows up in run_2's recorded topic list. Adding them to the bag recorder's topic allowlist lets us run the same gravity / orientation / rate audit retroactively on future bags without additional instrumentation.

8. What we explicitly do NOT recommend

Switching GICP's imu_topic to a gravity-removed or INS-attitude-fused IMU (such as VectorNav's ins_group-derived linear acceleration, or any downstream "corrected" IMU that absorbs INS attitude into orientation). Reasons:

(a) DLIO's math is built around specific force. Removing gravity in integrateImu + propagateState + the stationary calibration is non-trivial and creates new failure modes (e.g., vehicle pitched on grid while gravity removal is based on stale attitude).

(b) INS-fused linear acceleration depends on the vendor's attitude estimate. Errors there bleed into GICP's prior. With raw specific force, the only attitude in play is the one GICP itself maintains.

(c) Vendor INS-fused IMU is lower-rate than raw, hurting deskew.

(d) The current bias compromise already handles the dominant concern (online drift-chasing) without any code surgery.

9. Appendix — how to reproduce

The empirical numbers were generated from run_2_0.mcap with a Python script using mcap-ros2-support. To re-run:

```
pip install --user mcap mcap-ros2-support
PYTHONPATH=$HOME/.local/lib/python3.10/site-packages \
  python3 inspect_imu_all.py
# Topics inspected:
#   /gps_na/imu (NovAtel, 198 Hz, novatel_a, raw RAWIMUSX)
#   /gps_p1/imu (Atlas, 99 Hz, gps_antenna_top, imu_calibrated)
# Per topic the script prints mean/std of a_{x,y,z} and w_{x,y,z},
# orientation populated check, and a stationary-window summary.
```

race_common driver-side artifacts were inspected on branch stable at:

```
src/launch/iac_launch/launch/gps_launch/oem7_net_a.launch.py
src/launch/iac_launch/launch/gps_launch/atlas.launch.py
src/launch/iac_launch/launch/gps_launch/vectornav.launch.py
src/launch/iac_launch/param/gps_param/novatel_a/
  std_oem7_raw_msgs.yaml      # IDs requested from the receiver
  std_msg_topics.yaml          # IMU -> imu/data,
                               # RAWIMU -> imu/data_raw
  oem7_supported_imus.yaml     # IMU type -> rate map
src/launch/iac_launch/param/gps_param/vectornav/
  vectornav_interface.param.yaml   # imu.calibrated: true
```